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EVIDENCE FOR AN EARLY ARCHEAN GEOMAGNETIC FIELD: A PALEOMAGNETIC STUDY OF THE KOMATI FORMATION, BARBERTON GREENSTONE BELT, SOUTH AFRICA

C. J. Hale and D. J. Dunlop

Geophysics Laboratory, Departments of Geology and Physics
University of Toronto, Toronto, Canada M5S 1A7

Abstract. 3.5 Ga old peridotitic komatiites of the Komati formation, Barberton greenstone belt, South Africa have a characteristic NRM (natural remanent magnetization) that is stable during alternating field and thermal demagnetization. The thermal nature of this remanence is demonstrated by successful Thellier-type paleointensity determinations on some samples in the temperature range 350 to 575°C. The characteristic NRM is an almost total thermal overprint acquired during unroofing and cooling following pervasive greenschist metamorphism at 3470±20 Ma. The mean direction of magnetization after cleaning is D=326° I=81° (k=60, α_{95}=8°, N= 7 sites), with a paleopole at 11°S, 21°E (δp= δm=15°). The paleopole falls in southern Africa; ancestral southern Africa was therefore in polar latitudes around 3.5 Ga, the same position it occupied in the late Archean and early Proterozoic. Paleofield intensities are about one-half present-day intensities for polar latitudes. A vigorous earth's magnetic field was already in existence in the early Archean, implying that the core had a suitable size, composition and convective flow to support dynamo action.

Introduction

We report paleomagnetic directional data and paleofield intensities from the Komati formation, Onverwacht group, South Africa. These lavas erupted 3540±30 Ma ago (Sm/Nd, Hamilton et al., 1979) and were metamorphosed to greenschist facies shortly afterwards. In $^{40}Ar/^{39}Ar$ stepheating analyses of metamorphic mineral assemblages from our samples, Lopez-Martinez et al. (1984) have found plateau ages of 3450-3490 Ma. The characteristic magnetization in our study is an almost total thermal overprint dating from this metamorphism.

The Onverwacht volcanics are a rare example of a well-preserved volcano-stratigraphic sequence in which early Archean magnetization might survive. A generalized geological map after Viljoen and Viljoen (1969) is given in Fig. 1. The areal extent of the Onverwacht volcanics within the Barberton greenstone belt is shown in the inset.

The Komati formation consists of up to 3500 m of steeply dipping alternating mafic and ultramafic horizons, which include pillowed lavas as well as spinifex-textured peridotitic komatiite, believed to be a product of ultramafic extrusion. The ultramafic rocks have been almost totally serpentinized and have experienced regional greenschist-grade metamorphism as well as local amphibolite-grade contact metamorphism near the margins of tonalite intrusions.

The ultramafics tend to weather high, making them easy to sample but also resulting in extensive paleomagnetic contamination from modern lightning strikes (cf. Cisowski, 1981).

Serpentinization of original olivine grains has led to formation of micron- and submicron-size magnetite blebs, concentrated along grain boundaries. This fine-grained magnetite is the major carrier of paleomagnetic remanence.

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Sampling and Measurement Procedures

Ninety-six samples oriented by sun compass were collected from 16 sites in the Komati formation (Fig. 1). Two specimens of each sample were AF (alternating field) cleaned in steps to 100 mT. One specimen of each stably magnetized sample was then thermally demagnetized (as part of Thellier-type paleointensity determinations) in steps to 600°C.

Natural remanent magnetizations (NRM's) were measured on spinner magnetometers. As a result of serpentinization, the peridotitic samples have rather intense NRM's (typically 10 A m^{-1}), which are nevertheless <1% of saturation remanence. Most basaltic komatiites are very weakly magnetized (<10^{-3} A m^{-1}).

Characterization of Remanence Carriers

Thermomagnetic analysis of 15 representative samples revealed pure magnetite with essentially reversible behaviour. Some peridotites contained also a phase whose magnetization decreased irreversibly around 300°C. This mineral is likely maghemite, which was observed in thin section.

Koenigsberger Q_n ratios group between 1 and 10 with a peak around Q_n=3. Coercive force H_c varies from about 8 to >25 mT. The ratio of remanent coercive force H_{RC} to H_c is between 2 and 3 and the ratio of saturation remanence J_{rs} to saturation magnetization J_s is typically 0.2 but ranges up to 0.35. These properties indicate magnetite with a grain size less than a few microns.

Paleodirectional Results

A 3-tier analysis of directions of the characteristic component of NRM after cleaning is given in Table 1. In 38 samples from 7 sites, the NRM is a single-component steeply downward vector. This characteristic magnetization has a direction quite unlike that of the present earth's field (PEF) or the dipole field. It is directionally very stable in both AF and thermal demagnetization. Means and circles of

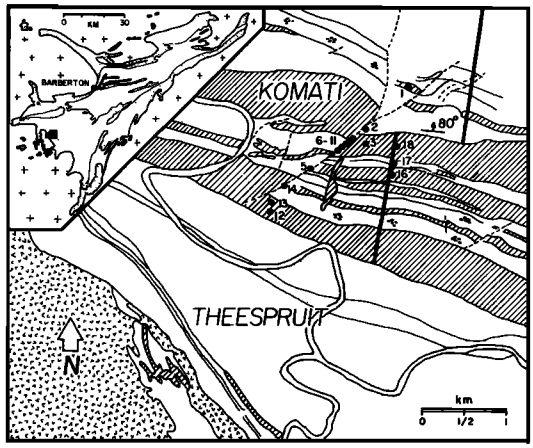


Fig. 1. A geological sketch map of part of the Komati and Theespruit Formations of the 3540 Ma Onverwacht Group and, on the lower left, a portion of the younger granitic Theespruit pluton. The numbered points are Komati sampling sites.

95% confidence for these sites are plotted in Fig. 2(a).

The characteristic direction was also revealed in 2-component NRM's of 5 samples from site 6 (Fig. 2(b)). The second, lower-coercivity NRM in site 6 samples has an east to northeast shallow direction. It is not found at other sites.

Paleointensity Results

Coe-modified Thellier paleointensity experiments (Coe, 1967) were carried out in vacuum on the samples that had stable single-component NRM's. Fig. 3 shows two successful results (KT7F12 and KT8F51) and an unsuccessful determination (B41D21). KT7F12 and KT8F51 have a characteristic form. At intermediate temperatures (300 or 350°C to 575°C), there is a linear trajectory incorporating 6-8 points representing about one-half the NRM. Our paleointensity estimates are based on the slopes of these trajectories. An initial steep descent between 200 and 350°C probably results from inversion of maghemite. The end product is hematite, which is much less magnetic than magnetite and has blocking temperatures outside the linear trajectory range. Above 575°C, there is a sharp increase in pTRM (partial thermoremanence), reflecting irreversible chemical changes. Such high-temperature alteration is commonly observed in Thellier experiments even though the heatings are carried out in vacuum.

The success of the Thellier experiment with those samples that did not alter continuously during heating demonstrates that the NRM has the characteristics of TRM. The implied paleofield intensities are comparable to those of the PEF. For example, KT7F12 and KT8F51 yield paleofield values of 34 μT and 36 μT respectively.

Discussion

How old is the characteristic remanence? We know that these lavas erupted underwater, some as pillow basalts, at about 3540 Ma (Hamilton et al., 1979). In the presence of water, rapid serpentinization would have resulted, like that seen in ultramafic plutonic rocks sampled by DSDP (e.g., Dunlop and Prévot, 1982). Because the NRM is apparently a TRM, we believe that the characteristic remanence was acquired later, as a nearly complete thermal overprint resulting from greenschist facies metamorphism. This metamorphism could have followed soon after extrusion, like the autometamorphism of the modern oceanic crust suggested by Stern et al. (1976).

Lopez-Martinez et al. (1984) have dated the greenschist metamorphism as having occurred within 100 Ma of extrusion, using $^{40}Ar/^{39}Ar$ results from metamorphic tremolites in 13 samples of our material. Most of their plateau ages are between 3.45 and 3.49 Ga. We therefore believe that our characteristic remanence is a thermal overprint rather than a truly primary TRM, and that its cooling age is about 3.47±.02 Ga. This study is one of the few instances in which $^{40}Ar/^{39}Ar$ dating and paleomagnetic work have been carried out on the same samples.

The Komati characteristic paleopole (Fig. 4) is notably distinct from most published Archean and early Proterozoic poles for southern Africa. No structural correction has been made in calculating the Komati pole, since we believe the characteristic is a post-kinematic remanence acquired during unroofing and cooling from ∿500°C. A contact test was unsuccessful because samples of the Theespruit pluton (Fig. 1) were unstably magnetized.

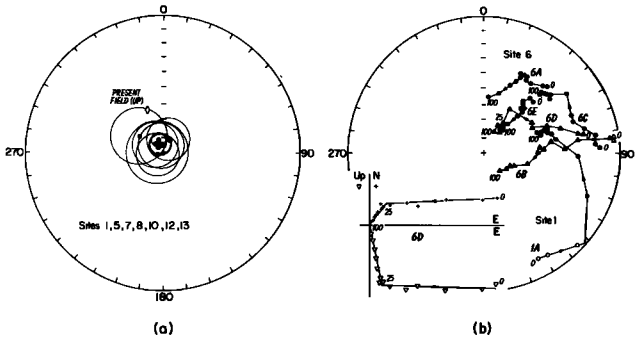
It would be futile to connect the few Archean poles of Fig. 4 so as to form an apparent polar wander path (APWP). The Komati pole is well dated but the same cannot be said of many other poles, and there are clearly long gaps in the record. For example, the Komati pole is the only result for the period 3.0-3.5 Ga, a time span almost as long as the Phanerozoic.

TABLE 1. Mean AF cleaned directions of the Komati characteristic remanence.

Site*	N	D(°)	I(°)	k	α_{95}
1	5	224	86	38	12
5	5	309	88	14	21
7	8	348	83	110	5
8	5	326	81	30	14
10	5	302	71	20	18
12	5	2	84	46	11
13	5	340	80	63	10
Mean	7	326	81	60	8
Paleopole		21E	11S	(δp= δm=15°)	

*Samples from sites 6 and 11 are excluded because they have bivectorial magnetizations; sites 3, 16 and 18 are lightning struck; sites 2, 9, 14 and 17 are in weakly magnetic basaltic komatiites; and sites 4 and 15 are from younger dikes.

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(a) (b)

Fig. 2. (a) Site-mean directions for the characteristic NRM component of the Komati Formation. The grand mean is shown as a star. (b) Two-component magnetizations of samples from site 6, illustrated on a stereoplot and an orthogonal vector diagram. The low-coercivity component is removed in AF's of 0-25 mT and the high-coercivity characteristic NRM is erased from 25-100 mT.

The poles of Fig. 4 cluster in a small region of the globe. They seem to record little or no apparent polar wander in the Archean. We have no comparable Archean records from other continents, but Proterozoic APWPs tend to consist of long tracks and "hairpins" (Irving and Park, 1972), recording major continental displacements and rotations. If the data of Fig. 4 are representative, Archean continental drift of southern Africa must have been quite minor. One could further speculate that plate tectonics did not operate in the Archean as it does now. The Kaapvaal craton was near the geographic pole at ∿3.5 Ga and seems to have remained there until ∿2.5 Ga.

We have used the geomagnetic field at 3.5 Ga (assuming it was dipolar) to pinpoint the position of ancestral southern Africa. Of equal importance is the very existence of a vigorous

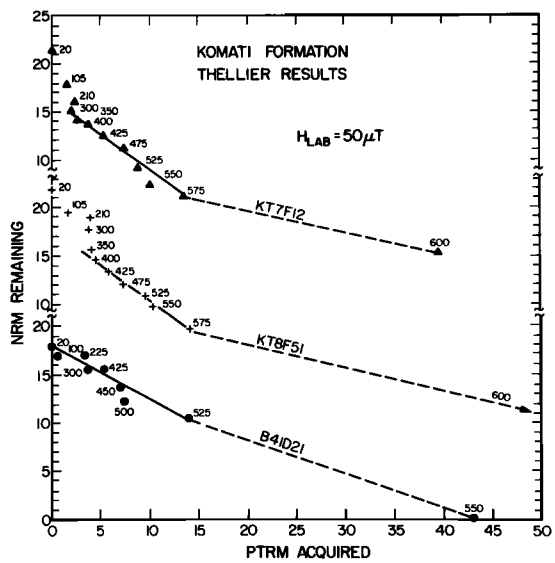


Fig. 3. Examples of Thellier-type paleointensity determinations for Komati samples. The numbers on the graphs are temperatures in °C.

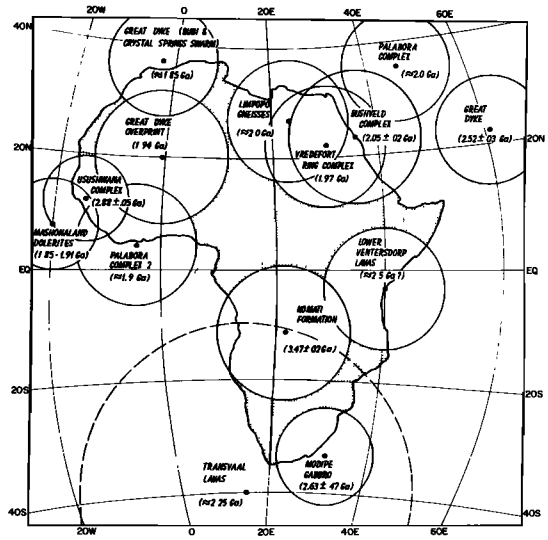


Fig. 4. Archean and early Proterozoic paleopoles for the Kaapvaal and Zimbabwe cratons of southern Africa, after Brock (1981), with the addition of poles for the Usushwana Complex (Layer et al., 1983) and the Komati Formation (this study).

magnetic field at 3.5 Ga, comparable in its strength to today's field. This is the first successful determination of paleofield intensity for early Archean terrestrial rocks; all previous determinations for rocks this old have been on lunar materials (cf. Cisowski et al., 1983). The earth had a relatively strong magnetic field very early in its history. The earth's core must therefore have been large enough and convecting vigorously enough to support dynamo action at this time.

These conclusions are of course speculative, based as they are on results for a single craton at a single time. Unfortunately our knowledge is unlikely to expand dramatically with further study because there are very few well-preserved early Archean rocks on earth that have not suffered pervasive late Archean or Proterozoic metamorphism. Rocks from Isua, Greenland, for example, have unreset Rb/Sr ages of 3.75 Ga (Moorbath, 1980) but their magnetization seems to have been updated around 1.7 Ga (Fahrig and Bridgwater, 1976; Beckmann, 1979). Only McElhinny and Senanayake's (1980) paleodirectional results from the Australian Pilbara craton are perhaps of comparable age to ours.

Conclusions

1. The earth had a magnetic field almost 3.5 Ga ago that was similar in strength to today's field. If the field was global, the earth's core must have had an appropriate size, composition and convective pattern at this stage in its evolution to support dynamo action.

2. Assuming the field was dipolar, ancestral southern Africa was in polar latitudes around 3.5 Ga ago, the same position it occupied in the late Archean and early Proterozoic.

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the field. Support through an Ontario Graduate Fellowship and a GSA Penrose Grant to CJH and NSERC Grant A7709 to D. J. Dunlop is gratefully acknowledged.

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